PROJECT PROPOSAL

Global Cholera Trends: A Statistical Analysis

PROBLEM STATEMENT

Despite global initiatives aiming to reduce cholera mortality to below 1%, the disease remains a significant public health threat. This project seeks to uncover the patterns and trend of the cholera disease over time and to answer the question: Is there a measurable decline in the severity of outbreaks over time and does this decline vary significantly by WHO region?

WHY IS THIS PROBLEM IMPORTANT

Cholera is a disease of inequity, acting as a marker of poverty, lack of social development, and inadequate access to basic services. Understanding where the disease is most lethal allows for the strategic allocation of Oral Cholera Vaccines (OCV) and Water, Sanitation, and Hygiene (WASH) resources. By analysing this dataset, we can identify the most vulnerable populations, the regions that require more aggressive public health interventions

DESCRIPTION OF DATASOURCE

The dataset is a WHO cholera dataset (sourced from Kaggle), covering global cholera surveillance data from 1949 to 2016.

Key Variables of Interest:

- Year
- Country
- WHO Region
- No. of Cases; how quickly the disease is spreading(incidence)
- No. of Deaths; the total burden
- Case Fatality Rate (CFR); severity of an outbreak

ANALYTICAL OBJECTIVES

This analysis aims to

- Baseline Characterization**:** Establish the cumulative global burden and identify "outlier" years of extreme epidemic activity.
- Trend Assessment: Determine if the global volume of cases (and severity?) is increasing, decreasing, or stagnating over the 67-year period.
- Hypothesis Testing: Statistically compare the Case Fatality Rate (CFR) between certain regions to identify disparities in outbreak lethality.

PROPOSED METRICS

- Mortality (Total Deaths)
- Prevalence (Total Cases)
- Time Trend (Year)
- Geographic Location (Country/Region)
- Severity Indicator (Case Fatality Rate)
- Descriptive Statistics**:** Cumulative totals, Medians, and Interquartile Ranges (IQR).
- Trend Analysis
- Inferential Testing.

PROPOSED STATISTICAL METHODS

- Descriptive Statistics
- Time Series Analysis
- Inferential Statistics

RISKS AND CONSTRAINTS

- Historical under-reporting: there is a risk that older records are not as accurate as modern ones in certain regions may bias results.
- Geographical reporting standards: Reporting standards vary greatly by region. Some countries might report every single case, while others might only report cases that reach a hospital.
- Data Skewness: public health data for diseases like cholera frequently tend to have a skewed distribution. We expect to see a few years with massive outbreaks and some with few cases.
- Lack of demographic detail: The dataset provides totals counts but does not break down the data by age, gender, or specific location within a country. This is a constraint because we can identify *where* and *when* an outbreak happened, but we cannot see *who* was most affected, which limits specific healthcare recommendations.